

Typed Evaluation Models and the Collapse Between Conflict and Ignorance: A Case Study on Jev

Modelos de evaluación tipada y el colapso entre conflicto e ignorancia: un estudio de caso sobre Jev

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Abstract. Chat-based LLMs return free text that must be interpreted; a category of tools that emerged in September 2026, typed evaluation models ("System One models"), instead return a typed object directly usable by software. Jev (TypeSafe AI, reported by the vendor as launched on September 15, 2026) is the first commercial example of this category. This work situates it within a five-category landscape of AI model interaction and tests, with a minimal experiment, a prediction derived from annotated decision theory: that an output collapsed to a single calibrated probability cannot distinguish genuine conflict from genuine ignorance. For Jev's simplest question type (boolean/Noul), the experiment is consistent with that prediction: conflict cases (probability 0.50-0.57) and ignorance cases (0.46-0.48) fall within the same narrow band. But a second experiment, using the same model's Choice type with a schema that explicitly names "conflicting evidence" and "insufficient evidence" as options, separates both cases with probability 1.0 in all four completed cases. A third experiment, with Choice restricted to the two original options (no escape categories), reproduces neither the collapse nor the clean separation: an unanticipated directional bias appears, possibly tied to lexical cues in the input text. The central finding is therefore not that Jev collapses per se, but that the same model preserves, destroys, or unpredictably distorts the distinction depending on the question type and declared schema — an interface design risk more complex than a single follow-up experiment could anticipate.

Keywords: typed evaluation models; System One models; LLM-as-judge; annotated paraconsistent logic; representational collapse; typed abstention; Jev.

Resumen. Los modelos de chat basados en LLM devuelven texto libre que debe interpretarse; una categoría de herramientas que emergió en septiembre de 2026, los modelos de evaluación tipada ("System One models"), devuelven en su lugar un objeto tipado directamente utilizable por software. Jev (TypeSafe AI, reportado por el fabricante como lanzado el 15 de septiembre de 2026) es el primer ejemplo comercial de esta categoría. Este trabajo lo ubica dentro de un panorama de cinco categorías de interacción con modelos de IA y examina, con un experimento mínimo, una predicción derivada de la teoría de decisión anotada: que una salida colapsada a una sola probabilidad calibrada no puede distinguir conflicto genuino de ignorancia genuina. Para el tipo de pregunta más simple de Jev (boolean/Noul), el experimento es consistente con esa predicción. Un segundo experimento, con el tipo Choice y un esquema enriquecido, separa ambos casos con probabilidad 1,0. Un tercer experimento, con Choice restringido a las opciones originales, no reproduce ni el colapso ni la separación perfecta: aparece un sesgo direccional no anticipado. El hallazgo central no es que Jev colapse per se, sino que el mismo modelo preserva, destruye, o distorsiona la distinción según el tipo de pregunta y el esquema declarado.

Palabras clave: modelos de evaluación tipada; System One models; LLM-as-judge; lógica paraconsistente anotada; colapso representacional; abstención tipada; Jev.

1 Introduction

A conventional chat model, faced with the question "was the refund approved?", might answer "Yes, it seems so, although I'm not entirely sure..." — an unstructured string of text that the consuming software must re-parse or interpret, with no explicit score and no format guarantee. A category of tools that emerged in September 2026, typed evaluation models, inverts that relationship: the model is given a state (the available context or evidence) and a schema of questions, each with a declared type, and the model returns, for each question, a value of that type together with a calibrated probability.

TypeSafe AI calls its first model of this kind "System One" — the fast, intuitive counterpart to the slow, deliberate reasoning ("System Two") pursued by frontier LLMs [1]. According to the vendor, Jev does not generate text: it selects among the options of a schema supplied by the caller, so returning a malformed value or one outside the declared set would be structurally impossible [2] — a format guarantee, not a guarantee of content correctness. The vendor reports 20 to 200 times more speed and up to 400 times lower cost than a frontier LLM for this class of tasks [3], and a proprietary training method, Reinforcement Learning for Calibrated Decisions

(RLCD), aimed at explicit calibration [4]. None of these figures were independently audited in this work; they are reported as vendor claims, not as verified facts.

This work does not evaluate Jev's performance on its intended tasks. Before examining it as a case study (§3-§6), §2 situates it within a broader landscape of AI model interaction categories. A representational question is then examined — what does a model that collapses evidence to a single calibrated probability lose? — and, following an unexpected finding in the experiment itself, that question is refined into a more precise one: is the loss a property of the model, or of the type of interface chosen to query it?

2 Landscape of AI Model Interaction Categories

The binary opposition "free chat vs. typed" oversimplifies the real landscape. Figure 1 positions five categories along two axes: computational cost and rigidity/timing of the decision schema's fixation — not "input flexibility," a formulation from an earlier version of this figure that led to a misreading (it suggested that a generative chat model was less flexible than a typed model, when it is exactly the opposite in terms of what it can be asked). The correct axis runs from a schema fixed at training time (category 4, the most rigid) to a schema declared at each call (categories 2, 3, and 5), to the total absence of a schema (category 1, free text — the maximum possible expressive freedom, but with zero structural guarantee on the response).

1. Generative chat models — free text, no guarantee of format or type. This is the no-schema extreme of the axis: it accepts any question, but offers no guarantee about the shape of the answer.

2. LLM with forced output (constrained decoding) — still a general-purpose generative model underneath, but at each generation step the space of valid tokens is restricted according to a declared grammar or JSON schema [12]. The cost remains that of a full LLM.

3. LLM-as-judge — a frontier or fine-tuned LLM evaluates according to a natural-language criterion. When fine-tuned to a specific task, the literature documents that it can degenerate into a closed-domain classifier, with a performance drop outside that domain [13].

4. Classical discriminative classifiers — dedicated non-generative architectures (e.g., a fine-tuned BERT, or an evidential head such as EDL), deterministic, with a class schema fixed at training time. Reported as substantially cheaper and faster than an LLM-based judge, although the exact figure varies by source and is not taken here as precise data.

5. Jev and its peers (typed evaluation models) — the case study of this work.

Note on this taxonomy: it is not a strict partition. Categories 2 and 3 can overlap in practice — an LLM-based judge can use forced output to guarantee a valid JSON verdict — and are presented as heuristic axes (use, decoding technique, architecture, cost) rather than as mutually exclusive classes. It is the author's own synthesis of secondary sources, not a systematic review of the constrained-decoding and LLM-as-judge literature.

With that caveat, Jev sits at the convergence of categories 3 and 4: it inherits from category 3 the flexibility of a question schema declared at call time, and from category 4 the low cost and the guarantee of typed output by construction. The LLM evaluation ecosystem is converging toward two-tier architectures in 2026 — a cheap classifier that resolves most cases and escalates to a frontier judge only when the case is ambiguous [14] — which is, structurally, the same cascade that a typed-abstention taxonomy already formalizes for decision-making under insufficient or disputed evidence (the author's own unpublished research-program synthesis; see note in §7).

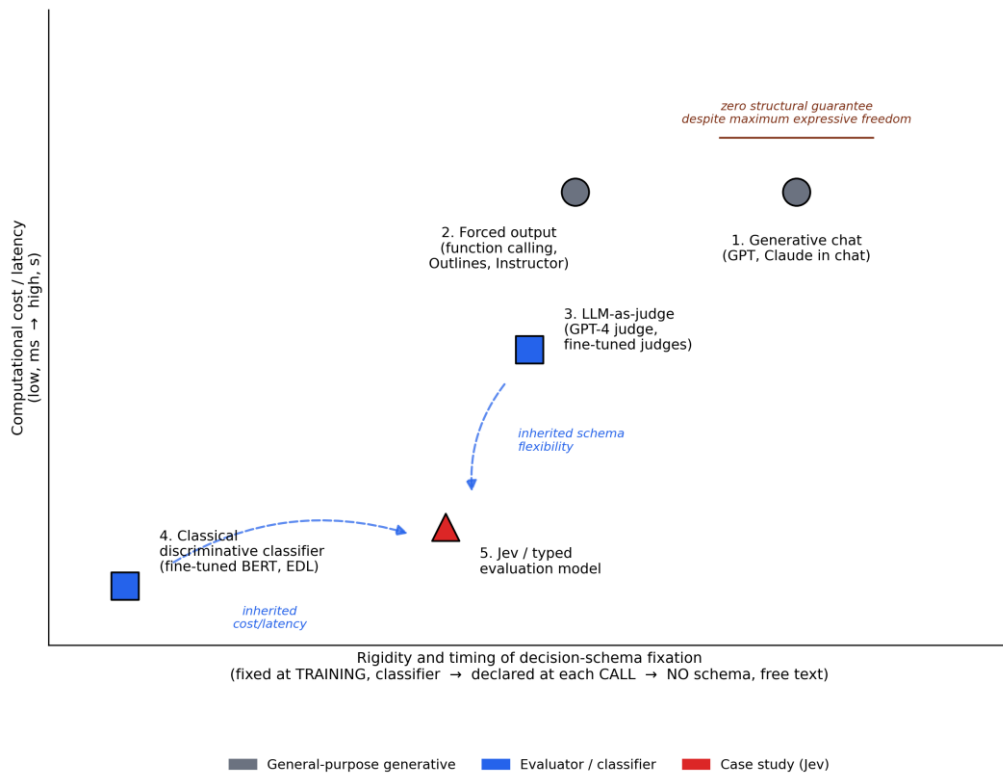


Figure 1: Positioning of the five AI model interaction categories according to schema rigidity/timing (x-axis: fixed at training → declared at each call → no schema) and computational cost (y-axis). Jev sits at the convergence between the LLM-based judge (category 3) and the classical discriminative classifier (category 4); generative chat (category 1) occupies the no-schema extreme — maximum expressive freedom, zero structural guarantee. Source: own elaboration.

3 Verified Taxonomy of Types in Jev

The vendor's documentation [8] specifies three question primitives, evaluable in parallel within a single call.

Table 1: Question primitives supported by Jev.

Type	What it does	What it returns
Choice	pick one option from a declared set (criteria: option→description map)	chosen option + probability per option
Score	rate the state on a rubric of ordered levels	score + probability per level
Noul	yes/no question	the probability that the answer is "yes"

Source: own elaboration.

The documentation calls the third type "Noul," not "Boolean," and describes it as a continuous probability in $[0,1]$ — not a discrete value. The Vercel AI Gateway SDK, which exposes `experimental_evaluate` as the access interface, names that same type `boolean` in its API signature [9]. Verified directly against the response object (§5): the field `providerMetadata.typesafe.confidence` exists in the schema but arrives empty (`{}`) in all five cases run with the boolean type — there is, for that type, no additional confidence or evidence-composition field beyond the single probability.

4 The Representational Question

In the annotated paraconsistent logic framework (LPA2v) [10], a judgment is represented as an independent pair (μ, λ) . Two distinguished vertices of the resulting lattice, $\top$ (inconsistent: strong evidence in both directions) and $\perp$ (paracomplete: absence of evidence in either direction), are epistemologically opposite but share a geometric property: both are equally far from the "true" and "false" ideal vertices [11]. From this a result already formalized follows: every closeness score to those ideal vertices is invariant under the involution that swaps $\top$ and $\perp$ [11].

This work does not apply that theorem literally to Jev — doing so would require Jev to explicitly compute an independent pair (μ, λ) , which is undocumented. The connection is a motivating analogy, not a deductive consequence of the theorem. The argument that is deductively valid, and that this work puts to the test, is simpler: no map from an evidence space of two or more dimensions (support, opposition, magnitude, insufficiency) to a sin-

gle scalar in $[0,1]$ can be injective; two distinct evidence states can, by pure cardinality, collapse to the same number. A model that exposes only that scalar — like Jev's Noul type — cannot, in principle, guarantee that conflict and ignorance produce distinct values.

5 Method

Six states (state) were designed in English on the same domain fact (whether a traffic light was red), in four categories: TORN (genuine conflict, $n=2$, two equally credible sources that contradict each other), SILENT (genuine ignorance, $n=2$, documented absence of evidence), AGREE-SUPPORT (control, $n=1$, sources agreeing on "red"), and AGREE-REFUTE (control, $n=1$, sources agreeing on "not red").

Experiment 1 (boolean/Noul type): one call per state to typesafe-ai/jev via `experimental_evaluate` from the ai SDK v7.0.107, single question: "Was the traffic light red at the time in question?"

Experiment 2 (Choice type, follow-up): the same six states, a single `lightState` question of type choice with four explicit options: red, green, `conflicting_evidence` ("reliable sources disagree with each other about the color"), and `insufficient_evidence` ("there is no evidence available to determine the color"). This second experiment was designed after observing the results of Experiment 1, prompted by an adversarial review that flagged the need to test whether the collapse was a property of the model or of the question type — it is explicitly declared as not preregistered and data-motivated.

Experiment 3 (binary Choice, closing the open question): the same six states, the same `lightState` question of type choice, but with only the two original options: red and green, no escape categories. Tests whether the perfect separation of Experiment 2 depends on the schema being able to name conflict/insufficiency, or whether it is a general property of the Choice type independent of the schema. Also data-motivated, not preregistered.

Experiment 4 (Score type, exploratory): a fourth experiment, not preregistered and motivated by the prior results, tests whether a purely scalar but ordinal question type — Score, which returns a probability-weighted mean over a rubric of ordered levels, not a categorical choice — also preserves the distinction between conflict and absence of evidence, or collapses like the boolean type. The Score syntax is not explicitly documented in the public specification consulted; it was determined by trial and error against the SDK's validation messages, which require an ordered array of levels (criteria), not a map as in Choice. A five-level rubric on "evidential determinacy" was used (0 = very high confidence, 4 = no confidence at all), deliberately agnostic about the cause of the uncertainty, and six new states were constructed — three of nominally increasing conflict intensity and three of nominally increasing evidence-absence intensity — plus reusing AGREE-SUPPORT as an already-determined control anchor. The six new states are in Appendix A.

Experiment 5 (μ/λ composition, exploratory): motivated by an idea discussed after the first four experiments: can an independent (μ, λ) pair, as in annotated paraconsistent logic (LPA2v, §4), be reconstructed by composing two separate calls to Jev's simplest type, instead of requiring the model to compute it internally? For each of the six original states (Appendix A), two boolean questions were asked in the same call: μ = "Is there credible evidence supporting the claim that the traffic light was red?" and λ = "Is there credible evidence supporting the claim that the traffic light was NOT red (i.e., was green)?", without forcing their probabilities to sum to 1 — independent questions, not a choice among options. Exploratory, not preregistered, motivated by the prior results.

Experiment 6 (third indeterminacy coordinate, validated with an expanded dataset): motivated by whether a third indeterminacy coordinate (I), as in neutrosophic logic (T, I, F), captures something the (μ, λ) pair of Experiment 5 does not. A seventh state, AMBIGUOUS, was designed (evidence that is intrinsically vague from a single source — neither contradictory like TORN nor absent like SILENT), and, alongside μ and λ , I was asked: "Is it indeterminate whether [the claim] — that is, does the available evidence fail to clearly resolve this either way, regardless of the reason why?" A first single-call-per-category pilot ($n=1$, traffic-light domain) suggested I did not distinguish TORN, SILENT, or AMBIGUOUS from each other — all three fell in the same high band — but with $n=1$ that conclusion could not be trusted. An expanded dataset of 20 heterogeneous domains was then built (traffic light, defective product, workplace attendance, medical biopsy, bank authorization, weather, contract, inventory, security, academic submission, vehicle brakes, food-safety inspection, building foundation, insurance, election, software, customs, veterinary, aviation, real estate), each instantiating the same 5 categories (TORN, SILENT, AGREE-SUPPORT, AGREE-REFUTE, AMBIGUOUS) with an identical structural template across domains, for a total of 100 states ($n=20$ per category). The full domain list is in Appendix A. A first attempt with μ/λ questions phrased meta-referentially ("does it support the direction labeled true in the state?") detectably degraded the model's calibration — an AGREE-REFUTE control that should have given low μ /high λ instead gave values nearly identical to AGREE-SUPPORT — so μ/λ were reformulated as concrete, domain-specific claims, as in Experiment 5, and the full dataset was rerun before reporting results.

None of the six experiments averages repetitions within a single state (a single call per case; limitation in §8) — Experiment 6 gains statistical power by repeating the case structure across 20 domains, not by repeating the same case. The calls were made on September 19 and 20, 2026 against the production model, with no access to its weights or training set.

6 Results

Table 2: Experiment 1 (boolean/Noul type) — probability of "red".

Case	Category	P(red)
TORN-1	genuine conflict (contradictory witnesses)	0.57
TORN-2	genuine conflict (contradictory sensors)	0.50
SILENT-1	genuine ignorance (no witnesses or cameras)	0.46
SILENT-2	genuine ignorance (lost record)	0.48
AGREE-SUPPORT	control: sources agree (support)	0.83
AGREE-REFUTE	control: sources agree (refutation)	0.08

Source: own elaboration.

The two TORN cases (0.50–0.57) and the two SILENT cases (0.46–0.48) fall within a common, narrow band. Both controls, with unambiguous evidence, yield values far from 0.5 in the correct direction: 0.83 for support ("red" evidence) and 0.08 for refutation ("green" evidence).

Table 3: Experiment 2 (Choice type, enriched schema) — choice and probability.

Case	Category	Jev's choice	P(choice)
TORN-1	genuine conflict	conflicting_evidence	1.00
TORN-2	genuine conflict	conflicting_evidence	1.00
SILENT-1	genuine ignorance	insufficient_evidence	1.00
SILENT-2	genuine ignorance	insufficient_evidence	1.00
AGREE-SUPPORT	control: sources agree	red	1.00
AGREE-REFUTE	control: sources agree	green	1.00

Source: own elaboration.

With the Choice type and a schema that explicitly names "conflicting evidence" and "insufficient evidence" as first-class options, Jev separates the four experimental cases (TORN vs. SILENT) with maximum probability and no ambiguity, and correctly classifies both controls (support → red, refutation → green), both at maximum probability. The same textual evidence that collapsed to a 0.46–0.57 band under the boolean type produces a perfect distinction under the Choice type.

Table 4: Experiment 3 (binary Choice type, red/green, no escape categories).

Case	Category	P(red)	P(green)
TORN-1	genuine conflict	0.85	0.15
TORN-2	genuine conflict	0.84	0.16
SILENT-1	genuine ignorance	0.67	0.33
SILENT-2	genuine ignorance	0.77	0.23
AGREE-SUPPORT	control: sources agree	1.00	0.00
AGREE-REFUTE	control: sources agree	0.00	1.00

Source: own elaboration.

Without escape categories, Jev does not reproduce the ~0.5 collapse of Experiment 1 nor the perfect separation of Experiment 2. Instead, a directional bias toward red appears in TORN and SILENT (0.67–0.85), more pronounced in TORN (0.84–0.85) than in SILENT (0.67–0.77) — a weak ordinal trend (n=2 per category) that does not allow a firm conclusion. The refutation control (AGREE-REFUTE), with unambiguous green evidence, does not follow that bias: it is classified cleanly and correctly (P(red)=0.00), just as sharply as the support control in the opposite direction. The bias toward red is therefore not an indiscriminate model artifact — it appears only in cases with ambiguous (TORN) or absent (SILENT) evidence. It is a third pattern, unanticipated by either of the two simple hypotheses (symmetric collapse or clean separation).

Table 5: Experiment 4 (Score type, graded determinacy).

Case	Branch	Score (0-4)
ANCHOR-DETERMINED	control (sources agree)	0.25
CONFLICT-MODERATE	conflict, unequal witness vantage	1.58
CONFLICT-SEVERE	conflict, maximally reliable sensors	2.04
CONFLICT-MILD	conflict, self-doubting witnesses	2.67
SILENCE-MODERATE	absence, vague unreliable rumor	3.66
SILENCE-MILD	absence, evidence irrelevant to the fact	3.88
SILENCE-SEVERE	total absence of evidence	3.95

Source: own elaboration.

The clean contrast is at the extremes: CONFLICT-SEVERE (2.04, "moderate," with 0.97 probability concentrated on that level) and SILENCE-SEVERE (3.95, "no confidence," 0.97 probability on that level) are separated

by nearly two points on a five-level scale — the Score type, although it also returns a scalar, does not collapse severe conflict and total absence to the same value, unlike the boolean type (Experiment 1). The control anchor (AGREE-SUPPORT) confirms that the maximum-determinacy extreme works as expected (0.25, near "very high confidence").

The three intermediate points in each branch, however, do not order monotonically as intended by design: within conflict, CONFLICT-MODERATE (1.58) turns out more determinate than CONFLICT-SEVERE (2.04) and CONFLICT-MILD (2.67), reversing the intended severity order; within absence, SILENCE-MODERATE (3.66) turns out less indeterminate than SILENCE-MILD (3.88). On review, CONFLICT-MODERATE has an unintentional design asymmetry (one witness with a "reasonably clear view" versus one with a "partially obstructed view") that likely introduces a differential-credibility cue absent from the other two cases in that branch — that is, the "degree" manipulation was not well controlled, and the result does not support a claim that Score finely orders the degree of indeterminacy within a branch. These six intermediate cases are reported for transparency (Appendix C), but are not interpreted as evidence of monotonic tracking of degree.

Table 6: Experiment 5 (μ/λ composition, independent pair).

Case	Category	μ (supports red)	λ (supports not-red)	Vertex
TORN-1	genuine conflict	0.86	0.84	T (both high)
TORN-2	genuine conflict	0.64	0.62	T (both high)
SILENT-1	genuine ignorance	0.04	0.04	$\perp$ (both low)
SILENT-2	genuine ignorance	0.05	0.04	$\perp$ (both low)
AGREE-SUPPORT	control: support	0.91	0.06	"true"
AGREE-REFUTE	control: refutation	0.06	0.92	"false"

Source: own elaboration.

Composing two independent boolean calls — one for evidence in favor, one against, without forcing them to sum to 1 — reconstructs the four canonical vertices of LPA2v's lattice (§4), and separates TORN from SILENT more sharply than Experiment 1: both TORN cases fall in the both-values-high region (T, inconsistent) and both SILENT cases in the both-values-low region ($\perp$, paracomplete) — exactly the distinction a single boolean call could not make. The magnitude differs between TORN-1 (0.86/0.84) and TORN-2 (0.64/0.62); with $n=1$ per case it cannot be established whether that is a real signal or sample noise.

Table 7: Experiment 6 — 20-domain dataset ($n=20$ per category, mean $\pm$ SD).

Category	μ	λ	I
TORN	0.80 $\pm$ 0.04	0.84 $\pm$ 0.04	0.85 $\pm$ 0.02
SILENT	0.04 $\pm$ 0.01	0.10 $\pm$ 0.05	0.96 $\pm$ 0.01
AGREE-SUPPORT	0.91 $\pm$ 0.02	0.05 $\pm$ 0.01	0.13 $\pm$ 0.05
AGREE-REFUTE	0.04 $\pm$ 0.01	0.91 $\pm$ 0.02	0.12 $\pm$ 0.05
AMBIGUOUS	0.13 $\pm$ 0.04	0.15 $\pm$ 0.06	0.95 $\pm$ 0.02

Source: own elaboration.

The μ/λ pattern from Experiment 5 replicates with high consistency across the 20 domains: TORN clearly separates from {SILENT, AMBIGUOUS} (both low), and the controls land where expected. On I, three two-tailed Welch's t-tests show the three "undetermined" categories are statistically distinguishable from one another, though with very different magnitudes: TORN vs. SILENT (mean difference 0.109; $t=-25.80$; $df=26.7$), TORN vs. AMBIGUOUS (difference 0.096; $t=-17.67$; $df=38.0$), and SILENT vs. AMBIGUOUS (difference 0.014; $t=3.22$; $df=26.8$; $p\approx 0.003$). The first two are large, highly robust differences; the third is real but an order of magnitude smaller, and its magnitude (0.014) is comparable to the within-category standard deviation (0.008–0.017) — significant at the population level, but insufficient to confidently classify a single new case.

7 Discussion

The pattern in Experiment 1 is consistent with the prediction in §4: under the Noul type, Jev does not distinguish why the evidence is unclear. But Experiment 2 forces a correction of the interpretation: the loss is not a property of the Jev model, but of the question type and schema declared to query it. The boolean type forces, by definition, an output in a one-degree-of-freedom space — there is no way for it to return "conflict" or "insufficiency" even if the model internally distinguishes them. The Choice type, when the schema includes those categories as named options, does have representational room to express them, and the model exploits it with a sharpness that was not anticipated (probability 1.0, not a diffuse tendency).

This reframes the contribution of the work: it is not "Jev has an inherent representational limitation," but rather that the design of the query schema determines whether the distinction survives, and the simplest, cheapest type to use (Noul, a single yes/no question) is precisely the one that destroys it by construction. This is a usage risk, not a product engineering defect.

Experiment 3 answers the open question from v0.2 only partially, and in a more uncomfortable way than either of the two simple answers that were anticipated. The hypothesis "Choice always separates, regardless of the

schema" is ruled out: without escape categories, there is no clean separation (0.84 is not 1.00, and SILENT does not approach 0.5 from the other side). But the alternative hypothesis "without escape categories, Choice collapses just like Noul" is also ruled out: there is no symmetric collapse toward 0.5 in any case — all values, including the SILENT cases with no evidence whatsoever, lean toward red.

The most honest explanation available with this data, without over-interpreting an $n=2$ per cell, is a possible lexical confound in the design of the stimulus itself: the TORN and AGREE-SUPPORT states literally contain the word "red" in the text (someone asserts it, even if it is contested), while the SILENT states do not mention either "red" or "green" at all. If the model partly weighs the surface presence of the word naming an option — and not only whether that assertion is credible or disputed — that would produce exactly the observed pattern. This explanation is plausible and testable, but this work did not test it in a controlled way — it remains a genuine open question, not a closed finding.

The AGREE-REFUTE case, completed in a follow-up call after resolving the rate limit (§8), offers an additional — uncontrolled, but informative — test of that lexical hypothesis. Its text also literally contains the word "red" ("the traffic light was green, not red"), in an explicit negation clause. If the bias were a simple surface count of that word's occurrences, some pull toward red would be expected here too. Instead, Jev classifies this case with probability 1.00 toward green ($P(\text{red})=0.00$) — as clean as the support control's pull toward red. This does not rule out the lexical confound outright, but it does narrow its scope: the red bias in Experiment 3 is not a blind count of lexical occurrences, and it concentrates on cases with ambiguous or absent evidence.

What does hold up across the three experiments together: the deductive argument in §4 (no single scalar can be injective over a higher-dimensional evidence space) explains the collapse of the Noul type, but does not predict or explain the pattern of Experiment 3 — a Choice with two options is not a single injective scalar in the same way, and yet it failed to achieve the separation that its representational space would in principle allow. The practical recommendation stands, but becomes more cautious: to preserve the distinction between conflict and ignorance, avoiding the boolean type is not enough — the Choice schema must also explicitly name those states as first-class options, and it is worth empirically verifying that the model is not responding to superficial lexical cues in the state rather than to the actual structure of the evidence.

Experiment 4 adds a nuance to this explanation: the Score type also returns a scalar (a probability-weighted mean over ordered levels), and yet it clearly separates severe conflict from total absence of evidence (2.04 vs. 3.95). This suggests that the relevant variable is not scalar-vs-categorical, but what question the schema forces: the boolean type forces an answer about the first-order fact ("is it red?"), where conflict and absence are, by the argument in §4, indistinguishable in principle; the Score type, as used here, was declared over a second-order question ("how determinate is the evidence?"), which can, in principle, differentiate how much information there is from what that information says. The limitation is that this experiment did not manage (§6) to show that Score reliably orders intermediate degrees of indeterminacy, so the conclusion is limited to the extremes contrast, not a fine-grained scale.

Experiment 5 closes the loop opened in §4 in a way that was not anticipated: Jev does not need to compute an independent (μ, λ) pair internally for that pair to exist — it can be reconstructed by composing two calls to the simplest, cheapest type the model offers, one oriented for and one against, without forcing their outputs to sum to 1. That composition cleanly separates TORN from SILENT where a single boolean call (Experiment 1) could not, and recovers the four canonical vertices of LPA2v's lattice with higher fidelity than Choice's discrete approximation (4 labels, not a continuum) or Score's one-dimensional one. The limitation is the usual one: $n=1$ per case, no repetition; the magnitude difference between TORN-1 and TORN-2 cannot be attributed to a cause with this data; and whether this composition generalizes to domains other than the traffic light, or to evidence with more than two sources, remains an open question.

Experiment 6 corrects, with data, a hasty reading by the analysis itself: looking at the ranges of an $n=5$ -per-domain pilot, it was concluded that I "does not distinguish" SILENT from AMBIGUOUS. The Welch's t-test on the $n=20$ dataset shows it does distinguish them, though with a magnitude an order smaller than the one separating TORN from both. The lesson is twofold: first, I does carry information (μ, λ) alone do not expose — contrary to the pilot's first impression — which answers, with qualification, the motivating question of whether a third neutrosophic coordinate captures something genuine; second, that information has scientific value (it confirms the coordinate is not redundant) but limited operational value: the SILENT-AMBIGUOUS difference (0.014) is of the same order as the within-category standard deviation (0.008–0.017), so it does not allow confidently classifying a single new case from I alone. For anyone wanting to use I as a third coordinate, the honest recommendation is: it serves to corroborate at the aggregate level that the distinction exists, and is a useful component alongside (μ, λ) for separating conflict from non-conflict; it does not serve, with this design, as a case-by-case classifier of why something is undetermined within the non-conflict region. Alternative phrasings of the I question, which might widen that separation, were not tested.

8 Limitations

(a) [RESOLVED 2026-09-19] The AGREE-REFUTE control initially failed due to the Vercel AI Gateway free tier's rate limit in all three experiments, even with a payment method already on file — the provider requires

loaded paid credits, not just a card on file. That step was completed the same day (a \$20 credit purchase), and the case was run in an independent follow-up call; all three experiments now have $n=6/6$ complete cases; (b) a single call per case in each experiment, with no repetition to estimate variance — critical for Experiment 3, whose pattern ($n=2$ per category) does not allow distinguishing a real trend from sample noise; (c) Experiments 2 and 3 were motivated by prior results and were not preregistered — explicitly declared as such; (d) the lexical-bias explanation for Experiment 3 (§7) is plausible and received additional support from the AGREE-REFUTE case, but was not tested in a controlled way; (e) this is a four-day-old product at the time of writing — its behavior and limits may change without notice, and these results should be read as a snapshot of 2026-09-19; (f) the speed, cost, and training-method (RLCD) figures are vendor claims, not independently audited in this work; (g) the five-category taxonomy in §2 is the author's own synthesis, not a systematic review; (h) Experiment 4 (Score type, exploratory, not preregistered) shares limitations (b)-(c) above (a single call per case, no repetition, data-motivated), and in addition the intermediate-degree contrast within each branch turned out non-monotonic, with at least one identified design confound (vantage asymmetry in CONFLICT-MODERATE, §6); only the extremes contrast (severe conflict vs. severe absence) is reported as a finding, not a fine-grained degree scale; (i) Experiment 5 (μ/λ composition, exploratory, not preregistered) shares the single-call-per-case limitation; the magnitude difference between TORN-1 and TORN-2 (0.86/0.84 vs. 0.64/0.62) cannot be attributed to a cause without more repetitions; and whether the composition generalizes beyond the traffic-light domain, or to states with more than two evidence sources, was not tested; (j) Experiment 6, although validated with $n=20$ per category across 20 heterogeneous domains, shares the same structural template across domains (two reporting sources / total absence / one vague source) — it does not cover other forms of conflict or indeterminacy (numerical evidence, time series, more than two sources, partial rather than total disagreement); the I question was phrased only one way, without testing alternative phrasings that might widen the SILENT/AMBIGUOUS separation; and the author's own $n=1$ pilot led to a conclusion the expanded data corrected — a concrete reminder of why this work repeatedly declares the $n=1$ limitation in Experiments 1-5.

9 Conclusion

The first experiment in this work seemed to confirm that typed evaluation models inherit, by representational design, the collapse between conflict and ignorance already formally characterized in annotated decision theory. A second experiment showed that conclusion to be incomplete: the same model, with a Choice schema that explicitly names the disputed epistemic states, separates conflict from ignorance with maximum probability. A third experiment, designed to isolate whether that separation depended on the schema or on the question type, showed that reality is messier than either of those two stories: without escape categories, Jev neither collapses symmetrically nor separates cleanly, but instead exhibits an unanticipated directional bias, possibly tied to superficial lexical cues in the input text. A fourth, exploratory experiment, with the Score type, suggests that the relevant variable is not scalar-vs-categorical but what question each type forces — a scalar over "how determinate is the evidence" does separate severe conflict from total absence, though it did not manage to establish a fine-grained scale of intermediate degrees. A fifth experiment closed the loop opened by annotated decision theory in §4: composing two independent boolean calls — for and against, without forcing them to sum to 1 — reconstructs that theory's (μ, λ) pair without Jev computing it internally, and the lattice's four canonical vertices appeared where expected. A sixth experiment, validated on a dataset of 20 heterogeneous domains ($n=20$ per category), tested whether a third neutrosophic indeterminacy coordinate (I) captures something (μ, λ) do not: the answer is affirmative but qualified — I statistically distinguishes conflict from non-conflict (large effect) and, with smaller but still significant magnitude ($p \approx 0.003$), total absence of evidence from intrinsically vague evidence, though the latter distinction is too small relative to single-case noise to be useful case by case. The experiment's own process — a wrong conclusion at $n=1$, corrected at $n=20$ — illustrates in practice the sample-size limitation this work repeatedly declares.

The lesson that survives all three experiments is not about a fixed limit of Jev as a product, but about two distinct and equally real risks for anyone building on typed evaluation models: first, reducing an epistemically complex decision to the simplest available interface type (boolean/Noul) discards information by construction; second, even a more expressive interface (Choice) can fail in non-obvious ways — biased, not simply "less informative" — when the schema does not explicitly name the states that matter.

Conflict of Interest

The author declares no conflict of interest.

Data and Code Availability

Public repository: <https://github.com/mleyvaz/jev-typed-evaluation-collapse> — includes the script and results of Experiment 1 (run_experiment.mjs, results.json), Experiment 2 (run_experiment_choice.mjs, results_choice.json), Experiment 3 (run_experiment_choice_binary.mjs, results_choice_binary.json), the follow-up

call that completed the AGREE-REFUTE case (run_missing_refute.mjs), Experiment 4 (run_experiment_graded_score.mjs, results_graded_score.json), Experiment 5 (run_experiment_dual_noul.mjs, results_dual_noul.json), Experiment 6 (run_experiment_triple_mli.mjs + results_triple_mli.json, n=1 pilot; run_experiment_multidomain_mli.mjs + results_multidomain20_mli.json, 20-domain dataset), the Figure 1 script (make_fig1_taxonomy.py), and the adversarial review round that motivated the v0.1 → v0.2 → v0.3 revisions.

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Appendix A. Input States (verbatim)

The six states (state) used across the three experiments, identical in each — only the declared question (questions) changes.

TORN-1 (genuine conflict)

Witness A, a police officer with a clear view of the intersection, testified under oath that the traffic light was red at the moment of the collision. Witness B, a bystander standing next to Witness A with an equally clear view, testified under oath that the same traffic light was green at that exact moment. Both witnesses are considered reliable by the investigating officer; there is no indication either one is lying.

TORN-2 (genuine conflict)

Sensor 1, a calibrated high-precision traffic sensor, recorded the light as RED at 14:03:02. Sensor 2, an equally calibrated high-precision sensor mounted on the same pole, recorded the light as GREEN at the same timestamp, 14:03:02. Both sensors passed their most recent calibration check with no faults reported.

SILENT-1 (genuine ignorance)

No witnesses were present at the intersection at the time in question. No traffic cameras were operating in that area that day. There is no record, sensor log, or testimony of any kind describing the state of the traffic light at that moment.

SILENT-2 (genuine ignorance)

The traffic light's camera log for that day was permanently lost in a server failure before any backup was made. No witnesses have come forward. No other observation of the light exists in any form.

AGREE-SUPPORT (control, support)

Witness A testified that the traffic light was red at the time of the collision. Witness B, standing nearby with an independent line of sight, separately and independently confirmed that the light was red at that same moment.

AGREE-REFUTE (control, refutation)

Witness A testified that the traffic light was green, not red, at the time of the collision. Witness B, standing nearby with an independent line of sight, separately and independently confirmed that the light was green at that same moment.

The six new states for Experiment 4 (the seventh case, ANCHOR-DETERMINED, reuses the AGREE-SUPPORT text):

CONFLICT-MILD (conflict, mild)

Witness A said the traffic light "might have been red, but I couldn't say for sure — it happened so fast." Witness B said it "could have been green, though I only caught a glimpse." Neither witness is confident in their own account, and both readily admit they might be wrong.

CONFLICT-MODERATE (conflict, moderate)

Witness A, who had a reasonably clear view of the intersection, said the traffic light was red at the moment of the collision. Witness B, who was standing some distance away with a partially obstructed view, said the same traffic light was green at that moment. Both witnesses seem generally credible, though neither had an ideal vantage point.

CONFLICT-SEVERE (conflict, severe)

Sensor 1, a newly calibrated, redundant-triple-checked traffic sensor with a documented error rate of less than 0.001%, recorded the light as RED at 14:03:02.000, with full internal diagnostics confirming normal operation. Sensor 2, an independent sensor of the same specification mounted on the same pole and calibrated the same day, recorded the light as GREEN at the exact same timestamp, 14:03:02.000, with full internal diagnostics confirming normal operation. Both sensors are considered maximally reliable; there is no known explanation for the disagreement.

SILENCE-MILD (absence, mild)

A passing dashcam recorded a few frames of the intersection, but the traffic light itself is out of frame in all of them; only the road surface and nearby cars are visible. No other recording or testimony covers the moment in question.

SILENCE-MODERATE (absence, moderate)

A pedestrian who was not looking at the light mentioned, in an offhand and unprompted remark days later, that they "vaguely recall something about the light," but could not say what color when pressed, or whether they were even looking at the right intersection. No other information exists.

SILENCE-SEVERE (absence, severe)

No witnesses were present at the intersection at the time in question. No traffic cameras were operating in that area that day. There is no record, sensor log, or testimony of any kind describing the state of the traffic light at that moment.

Experiment 6 pilot's AMBIGUOUS state (intrinsically vague evidence, traffic-light domain):

A single grainy photograph shows the traffic light mid-transition, rendered as an indistinct orange-red blur due to motion blur and low light. The photo technician who examined the image says its quality makes it impossible to tell whether the light had already reached red or was still yellow. There is only this one photograph; no other witness, sensor, or recording of the moment exists.

The 20 domains of Experiment 6's expanded dataset.

Each domain instantiates the same 5-category template (TORN, SILENT, AGREE-SUPPORT, AGREE-REFUTE, AMBIGUOUS) used above for the traffic light, substituting the subject, the disputed claim, the sources, and the vague-evidence description. One full worked example (WEATHER) is shown, and for the remaining 19, the subject and disputed claim; the complete parameters are in `run_experiment_multidomain_mli.mjs` in the repository, generated programmatically by the `mkDomain()` function to avoid the wording confounds of Experiment 4.

Full example — WEATHER:

TORN: "Witness A, standing under the awning near the scene, reported that it was raining at the time of the incident. Witness B, standing at the same location with an equally credible basis, reported that it was NOT raining at the time of the incident. Both are considered reliable; there is no indication either one is mistaken."

SILENT: "No weather station reading was available regarding whether it was raining at the time of the incident. No dashcam or security footage from the area exists either. There is no record of any kind describing whether it was raining at the time of the incident."

AGREE-SUPPORT / AGREE-REFUTE: both sources agree it was raining / was not raining, worded in parallel to the TORN pattern.

AMBIGUOUS: "A single smartphone photo from a passerby shows wet-looking pavement, but the meteorologist who reviewed it says reflections from nearby sprinklers make it impossible to tell whether the wetness was from rain or irrigation. No other weather record exists."

The other 19 domains (subject → disputed claim):

Table 8: The other 19 domains of the Experiment 6 dataset.

Domain	Disputed claim
TRAFFIC	the traffic light was red
PRODUCT	the phone's screen was cracked upon arrival
ATTENDANCE	the employee was present at the 9am meeting
MEDICAL	the biopsy sample was malignant
FINANCE	the account holder authorized the wire transfer
CONTRACT	the contract clause was signed by both parties

INVENTORY	the warehouse had the item in stock on that date
SECURITY	the door was locked at closing time
ACADEMIC	the student submitted the assignment before the deadline
VEHICLE	the car's brakes were functioning properly before the accident
FOODSAFETY	the restaurant kitchen passed the health inspection that day
CONSTRUCTION	the building's foundation met code specifications
INSURANCE	the claimant's injury occurred on company property
ELECTION	the ballot was received before the polling deadline
SOFTWARE	the deployed code included the security patch
CUSTOMS	the shipment declared its full contents accurately
VETERINARY	the animal showed symptoms of the disease before treatment
AVIATION	the aircraft's pre-flight check was completed
REALESTATE	the property disclosure included the known defect

Source: own elaboration.

Appendix B. Call Code, per Experiment

Experiment 1 (boolean/Noul):

```
import { experimental_evaluate as evaluate } from 'ai';

const result = await evaluate({
  model: 'typesafe-ai/jev',
  state: STATE_TEXT, // one of the six states in Appendix A
  questions: {
    wasRed: {
      type: 'boolean',
      instructions: 'Was the traffic light red at the time in question?',
    },
  },
});
```

Experiment 2 (Choice, enriched schema):

```
const result = await evaluate({
  model: 'typesafe-ai/jev',
  state: STATE_TEXT,
  questions: {
    lightState: {
      type: 'choice',
      instructions: 'What was the state of the traffic light at the time in question?',
      criteria: {
        red: 'The evidence indicates the light was red.',
        green: 'The evidence indicates the light was green (not red).',
        conflicting_evidence: 'Reliable sources disagree with each other about the color.',
        insufficient_evidence: 'There is no evidence available to determine the color.',
      },
    },
  },
});
```

Experiment 3 (binary Choice, no escape categories):

```
const result = await evaluate({
  model: 'typesafe-ai/jev',
  state: STATE_TEXT,
  questions: {
    lightState: {
      type: 'choice',
      instructions: 'What was the state of the traffic light at the time in question?',
      criteria: {
        red: 'The evidence indicates the light was red.',
        green: 'The evidence indicates the light was green (not red).',
      },
    },
  },
});
```

Experiment 4 (Score, graded determinacy):

```
const result = await evaluate({
  model: 'typesafe-ai/jev',
  state: STATE_TEXT,
  questions: {
    determinacy: {
      type: 'score',
      instructions: 'How determinate (certain) is the available evidence about the color of
the traffic light, regardless of which color it points to?',
      criteria: [
        'Very high confidence: the evidence clearly and unambiguously points to one col-
or.',
        'High confidence: the evidence leans clearly to one color, with minor uncertain-
ty.',
        'Moderate: there is no reliable basis to favor one color over the other.',
        'Low confidence: there is almost no basis to judge the color.',
        'No confidence: there is no basis whatsoever to judge the color.',
      ], // ordered array, not a map
    },
  },
});
```

Experiment 5 (μ/λ composition, two independent boolean questions):

```
const result = await evaluate({
  model: 'typesafe-ai/jev',
  state: STATE_TEXT, // one of the six states in Appendix A
  questions: {
    mu: {
      type: 'boolean',
      instructions: 'Is there credible evidence supporting the claim that the traffic light
was red?',
    },
    lambda: {
      type: 'boolean',
      instructions: 'Is there credible evidence supporting the claim that the traffic light
was NOT red (i.e., was green)?',
    },
  },
});
// mu and lambda are not forced to sum to 1: independent questions, not a choice.
```

Experiment 6 (third coordinate I, 20-domain dataset):

```
function mkDomain(id, subject, claimTrue, claimFalse, reliableA, reliableB,
  primaryEvidence, secondaryEvidence, ambiguousEvidence) {
  const mu = { type: 'boolean',
    instructions: `Is there credible evidence supporting the claim that ${claimTrue}?` };
  const lambda = { type: 'boolean',
    instructions: `Is there credible evidence supporting the claim that ${claimFalse}?` };
  const indeterminacy = { type: 'boolean',
    instructions: `Is it indeterminate whether ${claimTrue} – that is, does the available `
      + `evidence fail to clearly resolve this either way, regardless of the reason why?`
  };
  return [ /* TORN, SILENT, AGREE-SUPPORT, AGREE-REFUTE, AMBIGUOUS, each with a state built
    from the template and {mu, lambda, indeterminacy} */ ];
}
// 20 calls to mkDomain(...) generate the 100 states; see
run_experiment_multidomain_mli.mjs
// in the repository for the 20 full parameter sets.

const result = await evaluate({
  model: 'typesafe-ai/jev',
  state: STATE_TEXT,
  questions: { mu: item.mu, lambda: item.lambda, indeterminacy: item.indeterminacy },
});
```

Appendix C. Full Response (answers) per Case

The answers object returned by Jev for each completed case, unedited (the full response object also includes usage, warnings, rounding, and providerMetadata, available in the results*.json files in the repository).

Experiment 1 — boolean/Noul:

```
TORN-1: {"wasRed":{"type":"boolean","probability":0.57}}
TORN-2: {"wasRed":{"type":"boolean","probability":0.50}}
```

```
SILENT-1:      {"wasRed":{"type":"boolean","probability":0.46}}
SILENT-2:      {"wasRed":{"type":"boolean","probability":0.48}}
AGREE-SUPPORT: {"wasRed":{"type":"boolean","probability":0.83}}
AGREE-REFUTE:  {"wasRed":{"type":"boolean","probability":0.08}}
```

Experiment 2 — enriched Choice:

```
TORN-1:      {"lightState":{"choice":"conflicting_evidence","probabilities":
  {"red":0,"green":0,"conflicting_evidence":1,"insufficient_evidence":0}}}
TORN-2:      {"lightState":{"choice":"conflicting_evidence","probabilities":
  {"red":0,"green":0,"conflicting_evidence":1,"insufficient_evidence":0}}}
SILENT-1:    {"lightState":{"choice":"insufficient_evidence","probabilities":
  {"red":0,"green":0,"conflicting_evidence":0,"insufficient_evidence":1}}}
SILENT-2:    {"lightState":{"choice":"insufficient_evidence","probabilities":
  {"red":0,"green":0,"conflicting_evidence":0,"insufficient_evidence":1}}}
AGREE-SUPPORT: {"lightState":{"choice":"red","probabilities":
  {"red":1,"green":0,"conflicting_evidence":0,"insufficient_evidence":0}}}
AGREE-REFUTE: {"lightState":{"choice":"green","probabilities":
  {"red":0,"green":1,"conflicting_evidence":0,"insufficient_evidence":0}}}
```

Experiment 3 — binary Choice:

```
TORN-1:      {"lightState":{"choice":"red","probabilities":{"red":0.85,"green":0.15}}}
TORN-2:      {"lightState":{"choice":"red","probabilities":{"red":0.84,"green":0.16}}}
SILENT-1:    {"lightState":{"choice":"red","probabilities":{"red":0.67,"green":0.33}}}
SILENT-2:    {"lightState":{"choice":"red","probabilities":{"red":0.77,"green":0.23}}}
AGREE-SUPPORT: {"lightState":{"choice":"red","probabilities":{"red":1,"green":0}}}
AGREE-REFUTE: {"lightState":{"choice":"green","probabilities":{"red":0,"green":1}}}
```

Experiment 4 — Score (graded determinacy):

```
ANCHOR-DETERMINED: {"determinacy":{"score":0.25,"probabilities":{"0":0.75,"1":0.25,"2":0,"3":0,"4":0}}}
CONFLICT-MILD:      {"determinacy":{"score":2.67,"probabilities":{"0":0,"1":0,"2":0.36,"3":0.62,"4":0.02}}}
CONFLICT-MODERATE:  {"determinacy":{"score":1.58,"probabilities":{"0":0,"1":0.43,"2":0.56,"3":0.01,"4":0}}}
CONFLICT-SEVERE:    {"determinacy":{"score":2.04,"probabilities":{"0":0,"1":0,"2":0.97,"3":0.02,"4":0.01}}}
SILENCE-MILD:       {"determinacy":{"score":3.88,"probabilities":{"0":0,"1":0,"2":0.03,"3":0.07,"4":0.90}}}
SILENCE-MODERATE:   {"determinacy":{"score":3.66,"probabilities":{"0":0,"1":0,"2":0.02,"3":0.30,"4":0.68}}}
SILENCE-SEVERE:     {"determinacy":{"score":3.95,"probabilities":{"0":0,"1":0,"2":0.02,"3":0.01,"4":0.97}}}
```

Experiment 5 — μ/λ composition:

```
TORN-1:      {"mu":{"probability":0.86},"lambda":{"probability":0.84}}
TORN-2:      {"mu":{"probability":0.64},"lambda":{"probability":0.62}}
SILENT-1:    {"mu":{"probability":0.04},"lambda":{"probability":0.04}}
SILENT-2:    {"mu":{"probability":0.05},"lambda":{"probability":0.04}}
AGREE-SUPPORT: {"mu":{"probability":0.91},"lambda":{"probability":0.06}}
AGREE-REFUTE: {"mu":{"probability":0.06},"lambda":{"probability":0.92}}
```

Experiment 6 — pilot (n=1, traffic-light domain, with I added):

```
TORN-1:      mu=0.87  lambda=0.84  I=0.89
TORN-2:      mu=0.65  lambda=0.60  I=0.88
SILENT-1:    mu=0.04  lambda=0.04  I=0.97
SILENT-2:    mu=0.05  lambda=0.04  I=0.95
AGREE-SUPPORT: mu=0.92  lambda=0.06  I=0.14
AGREE-REFUTE: mu=0.06  lambda=0.92  I=0.12
AMBIGUOUS:    mu=0.12  lambda=0.06  I=0.97
```

Experiment 6 — 20-domain dataset (n=20 per category). The full 100 rows are in `results_multidomain20_mli.json` in the repository; the per-category summary is in Table 7. Sample, WEATHER domain:

```
WEATHER-TORN:      mu=0.81  lambda=0.85  I=0.84
WEATHER-SILENT:    mu=0.03  lambda=0.05  I=0.97
WEATHER-AGREE-SUPPORT: mu=0.92  lambda=0.04  I=0.09
WEATHER-AGREE-REFUTE: mu=0.04  lambda=0.88  I=0.12
WEATHER-AMBIGUOUS: mu=0.13  lambda=0.14  I=0.95
```